

Martin Saulnier

French engineering student applying for Master's programs in Industrial Engineering in the United States

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EDUCATION

Arts et Métiers Institute of Technology (ENSAM)

Combined BS/MS Engineering Degree

Sept 2024 - Jul 2026

Paris, France

Arts et Métiers is a member of ParisTech, a consortium of prestigious French institutions recognized for academic excellence, outstanding faculty, and world-class research laboratories.

- Ranked top 8% (97th / 1150 Students) | **GPA 3.94/4.0**
- Mechanical, Electrical, Industrial Engineering, Design, Manufacturing, Fluid Dynamics, Thermodynamics, Material Science, Supply Chain.

Lycée Jules Haag

Intensive preparatory class, Mathematics, Physics, Industrial Engineering

Sept 2022 – Jul 2024

Besançon, France

A highly competitive two-year program in France, designed to prepare top students for entrance exams to elite engineering schools. Provides accelerated, university-level training that demands rigor, discipline, and advanced problem-solving skills.

- Elite track in Physics & Engineering Sciences | **GPA 4.0/4.0**
- Advanced Mathematics, Physics, Mechanical Engineering

SELECTED PROJECTS

Manufacturing Project — Shoulder Prosthesis Design and Optimization

Sep 2025 – Present

Team of 6 — Research Project (Biomechanics and Structural Optimization)

Objective: Develop and optimize glenoid components for shoulder prostheses to improve long-term implant stability and durability.

- Investigating implant loosening mechanisms (*Rocking-Horse effect*), identified as the leading cause of long-term prosthesis failure.
- Analyzing bone-implant contact mechanics and stability through structural and material modeling to identify key failure factors.
- Optimizing combinations of geometry, surface patterns, and materials to enhance fixation strength and long-term durability.

ENS Data Challenge — Enedis: Filling Missing Electricity Consumption

Sep 2025 – Present

Team of 4 — Research Project (Machine Learning and Time Series Analysis)

Objective: Reconstruct missing values in 1,000 electricity load curves using synthetic data and evaluated with MAE.

- Processing large-scale Linky-like datasets (140–260 MB, 21k–38k curves) to identify and complete missing values
- Building and testing models beyond linear interpolation, including seasonal decomposition, Kalman smoothing, and matrix completion
- Improving the benchmark MAE and developing solid skills in time-series ML, data engineering, and model evaluation.

AMJE Bordeaux — Internal App Development (React Native, Supabase)

May 2025 – Present

Professional Project (Mobile App Development for Student Consulting Club)

Objective: Design and deploy a cross-platform app centralizing studies, KPIs, and announcements for AMJE members.

- Developing the front-end with **Expo / React Native**, integrating real-time data synchronization via **Supabase**
- Implementing secure study creation, deletion, and status update flows with dynamic KPI and dashboard refresh
- Structuring the app for scalability and maintainability (modular TypeScript services, real-time subscriptions, reusable UI)

Python-Based Slack Bot for AMJE Bordeaux Consulting Club

Apr 2025

Professional Project (Automation and Workflow Optimization)

Objective: Streamline the study process by centralizing documents and introducing automated tools for tracking tasks and ensuring completion

- Created a system to aggregate and organize all relevant study documents, centralizing them in one accessible platform.
- Implemented automated checklists and task tracking to ensure all steps in the study process are completed efficiently.

UrbanWind : Smart Micro-Wind Turbine Transmission System

Sept 2024 - Jun 2025

Team of 5 — Professional Project with I2M Laboratory and ADV-Tech (Mechanical Design and Simulation)

Objective: Design and validate a compact transmission system for an urban wind turbine by combining analytical sizing with 3D modeling and simulations

- Designed and dimensioned a 4.9 kW urban micro wind turbine, from rotor blades to transmission shafts, using aerodynamic models and engineering catalogs.
- Engineered and validated a custom gearbox and bearing system with CAD (Catia V5) and finite element analysis (ABAQUS), ensuring long-term durability.
- Integrated all components into a manufacturable assembly, completing full CAD modeling and structural verification for 10+ years of continuous operation.

Research Project (Signal Processing and Acoustic Engineering)

Objective: Design and build a noise-filtering circuit based on lock-in amplification to suppress wind interference while maintaining voice clarity inside ski helmets

- Reproduced and demonstrated the use of synchronous detection with basic electronic components (filters, multiplier, function generator).
- Applied lock-in amplifier techniques to eliminate noise from the input signal, achieving a 70% reduction in background noise, resulting in a clean and audible output signal.
- Designed and miniaturized the electronic circuit to fit within a ski helmet for real-world deployment.

PROFESSIONAL EXPERIENCE

AMJE Paris

July 2025 – Present

Consultant, Student Consulting Club

- Contributing to engineering and business projects, delivering professional-quality work for client companies.

Complétude

June 2025 – Present

Private Tutor — Mathematics, Physics, Engineering Sciences

- Supporting students in math and science to improve results and challenge advanced learners (raising grades from B- to A)
- Building analytical skills and providing personalized guidance to foster academic progress

Arabelle Solutions (EDF Group)

June 2025 – July 2025

Nuclear Industry Blue Collar Internship

- Worked hands-on on the rotor assembly line of a world-class nuclear turbine manufacturer.
- Improved and organized Foreign Material Exclusion (FME) zones to enhance safety compliance.

AMJE Bordeaux

Jan 2025 – Present

Quality Manager, Student Consulting Club

- Leading the quality control process for ongoing research studies, ensuring compliance with AMJE standards and industry regulations.
- Conducting quality assessments (Mock Audits, reduced document processing time by 50%) and providing recommendations for process improvements, optimizing workflows and reducing errors.
- Training and mentoring new team members on quality control procedures and best practices for maintaining high-quality standards in research projects. (team of 14)

LEADERSHIP AND VOLUNTEERING

Réseau N+i

Sep 2025 – Present

Student Tutor

- Supporting the integration of 3 international students from China as part of the Réseau N+i program
- Providing weekly academic tutoring in STEM subjects, administrative guidance, and cultural immersion activities

Association des élèves des Arts et Métiers

May 2025 – Present

Student Tutoring Coordinator in Bordeaux

Association des élèves des Arts et Métiers is France’s largest student association with 3,500 members

- Organizing tutoring sessions and exam review lessons for first-year students
- Coordinating volunteer tutors and managing a shared repository of academic resources (+5000 documents)

GaSole

Sep 2024 – Present

Volunteer for the association

- Taking part in monthly Clean Walks and organizing sustainability workshops in primary schools
- Providing food and essential items to the homeless, offering moral support, and strengthening community ties

AWARDS AND ACHIEVEMENTS

Award for Outstanding Academic Achievement	2025
Ranked in the top 8% of Arts et Métiers first year students	2024
Gymnastics (Men’s Artistic) — 13 years at a Regional competitive level (FFGym Certification)	2017

SKILLS AND INTERESTS

- Languages: **French (Native)**, **English (TOEFL iBT : 110/120)**, **Spanish (Fully operational)**, **German (Intermediate)**
- Programming : Python, LaTeX, SQL, HTML & Java (Website)
- Software : 3DEXPERIENCE, Catia V5, SolidWorks, Abaqus, Simcenter STAR-CCM+;
- Sports and Leisure : **Soccer** (10 years with **competitions**), **Skiing** (8 years), **Gymnastics** , **Electric Guitar** (performing with my school’s band at events)
- Linguistics & Travel: Passionate about cultural studies. Studied English for 15 years, Spanish (mother tongue), Latin (3 years), Italian (1 year), and traveled to **8 countries**.